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Colin Tiernan

ALERT FEATURED

Grazing and grouse: Massive Idaho study hopes to explain impact of livestock on imperiled Western bird

Colin Tiernan

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Wildlife technicians Sam Siller, left, and Reese Wray weigh a sage grouse March 18, 2021, west of Rogerson.
COLIN TIERNAN, TIMES-NEWS FILE PHOTO

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ROGERSON — Reese Wray and Sam Siller are tearing across the desert in the middle of the night on an ATV.

The crescent moon's covered by clouds, so all you can see of them in the distance is their frantic spotlight, sweeping over the sagebrush in wide arcs. They pause from time to time, search, then hurtle around some more.

After about 15 minutes of looking in the sage-covered terrain near the Nevada border, Wray finds what he's looking for: The glint of turquoise eyes in the dusty beam of his light.



A male sage grouse sits calmly in wildlife technician Reese Wray's hands after being banded March 18 west of Rogerson. Sage grouse numbers continue trending downward across the West.

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Wray's hands start shaking involuntarily. He shimmies the light and speeds toward the bird. When they get to it, Siller hops off the ATV, runs at the petrified sage grouse and swings her net down on top of it. It's 11 p.m. on March 18 and they've got their first hen of the night. They'll tie a radio collar around her neck to track her movements, measure her, then let her go and continue spotlighting.

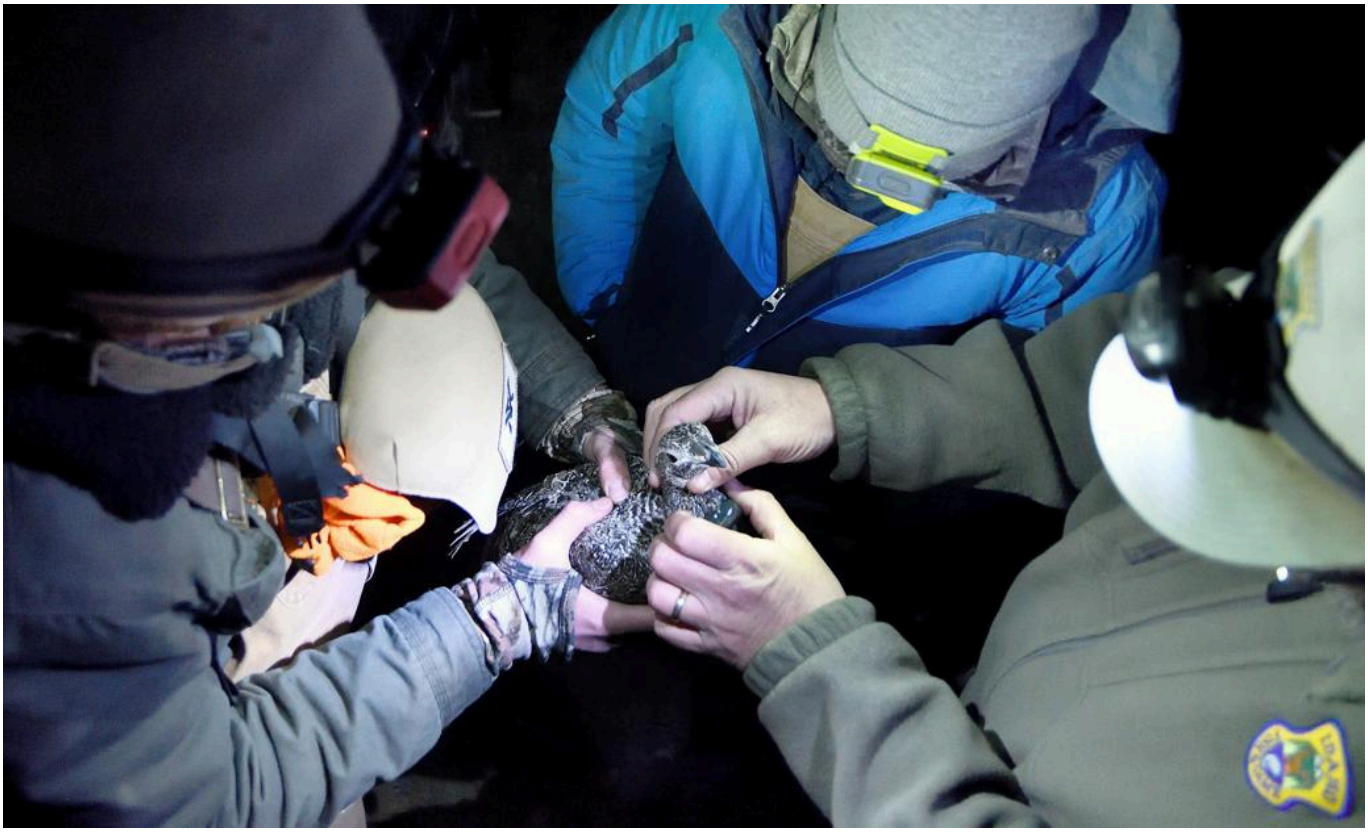
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From left, Reese Wray, Sam Siller and Dave Musil check that a sage grouse hen has been properly banded March 18 near Rogerson.

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There's a method to this madness. The night's sage grouse capturing and radio-collaring in no man's land near the Nevada border will help biologists answer one of the West's most important and emotional environmental questions: How does livestock grazing affect sage grouse?

It's not a new question and biologists have tried to answer it before. But, somewhat conspicuously, it hasn't been answered with a long-term research project like this, said Dave Musil, an Idaho Department of Fish and Game senior wildlife research biologist.

"There's never been a major experimental study with grazing," Musil said. "There's never been any controlled treatment, hasn't been any before and after. This project is new. It's groundbreaking."

Mega study

Musil could talk about sage grouse for hours on end.

One minute he'll be telling a story about a funny hen he radio-collared years ago; the next he'll explain the hyper-specific migration patterns of southern Idaho birds or how sage grouse have special feathers over their nostrils that help them breathe in winter.

He's been studying sage grouse for years. Back when he was catching birds full time, he'd insist on holding the spotlight because he didn't trust anyone else with it. He passed along some of his wisdom to the capture crew at 8:30 p.m. before they headed out: Don't start looking for birds too early.

"Let them be in their jammies," Musil said. "Let them get done reading their book and watching Johnny Carson and let them get sleepy. Let them get tired so they won't be so jumpy."



Senior wildlife research biologist Dave Musil holds a male sage grouse March 18 west of Rogerson. In his decades as a biologist and researcher with Fish and Game, Musil has never worked on a sage grouse project this large.

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Musil's clad in standard Fish and Game garb: green pants, gray shirt, sweat-stained baseball cap. He wears glasses and has a black beard under his mask.

He's at home in the sagebrush, nonchalantly driving an F-150 down absurdly bumpy two-tracks at night with the headlights off.

But in his decades as a biologist and researcher with Fish and Game, Musil has never done a project this big.

This research project is looking at the impacts of grazing on sage grouse over a 10-year period. It's unusual for a project to last that long.

“Most projects, (in) two to three years you get your answer,” Musil said. “That’s the most amazing part of this project, the long-term aspects that just blow other projects out of the water.”

The study began in 2014 and it’s a collaboration between a handful of groups and biologists — the University of Idaho, Fish and Game, U.S. Geological Survey and Bureau of Land Management are all involved on the research side.

“No one group could have done it,” Musil said. “It had to be a collaboration.”

In addition to those groups, the Idaho Cattle Association and federal grants are helping pay for the study. Ranchers are adjusting their grazing patterns so researchers can manipulate the study area.



A sage grouse hides in its natural sagebrush environment.

Mychel Matthews

It's an expensive undertaking, with the groups involved spending several hundreds of thousands of dollars every year to keep the project going. Roughly 40 field technicians and researchers run the captures, survey vegetation and monitor nest choice and success at the five sites.

The project is massive.

It spans five study sites, each of which could be reasonably described as in the middle of nowhere. Three sites lie south of the Snake River in Cassia, Twin Falls and Owyhee counties. One site is south of Salmon in the Pahsimeroi Valley and another is east of Craters of the Moon National Monument.

A sixth site at Idaho National Laboratory — which would have been unique because it hadn't been grazed in a half-century — burned in the 2019 Sheep Fire.

Researchers aim to have 50 radio-collared hens at each site at any given time. The batteries in the collars are usually good for two years, so you don't have to catch 50 birds every winter and spring, but it's still a challenge to get to that number.

A decade of data on hundreds of sage grouse would be useful on its own, but this study is unique because of how biologists are manipulating sage grouse nesting habitat.



Sage grouse look for food in a cheatgrass-infested part of the Idaho desert in 2020.

COLIN TIERNAN, TIMES-NEWS FILE PHOTO

Each study site has three treatments or grazing types. Part is ungrazed — that's the control. Another part is grazed every spring. The third treatment is an alternating grazing schedule: Grazed in the spring (on), rested in the fall (off), rested in the spring (off), grazed in the fall (on), then repeated.

Hens don't tend to stray too far from the spring lek areas where they're captured and collared, so the odds are good they'll nest in one of the treatments within the study area.

The question is, which treatment will they choose? And which one leads to the greatest nest success?

“The impetus of the study is to see: Does grazing have any positive or negative impact?” Musil said. “Are they (sage grouse) more likely to have success if it’s in a rested area? Or does it not even matter?”

Why do grouse and grazing matter?

Sage grouse aren’t in good shape.

In the 1800s, there may have been as many as 16 million of them throughout the West. The birds once lived from southern Alberta to northern Arizona, and from South Dakota to Oregon.



Male sage grouse display at dawn in 2020 at a lek near the Nevada border. Sage grouse are specially adapted to the sagebrush steppe. They lack a fully functional gizzard, but have a unique ability to digest sagebrush and excrete its volatile oils. In winter the birds will often feed on sagebrush.

COLIN TIERNAN, TIMES-NEWS FILE PHOTO

There might be a few hundred thousand or a half-million left in the sagebrush sea and the bird’s range has contracted by about 50%. Southern Idaho hunters say they simply don’t see many sage grouse anymore. A recent U.S. Geological Survey report

says the iconic bird's population has dropped 80% since 1965.

Sage grouse are chicken-sized birds that live in some of America's most rugged country. Outside of the breeding season, they don't look dramatic with their mottled brown, white and black plumage. They're best known for their elaborate mating rituals in late winter and spring.



Wildlife technician Reese Wray spreads the wing of a sage grouse to show its flight feathers, or "remiges," March 18, 2021, west of Rogerson.

COLIN TIERNAN, TIMES-NEWS FILE PHOTO

Males will gather at dawn and dusk at group breeding grounds, called leks. They'll fan their spiky tail feathers behind them like blue-collar peacocks, balloon the yellow air sacs on their chests and make a bouncy, alien, popping sound, all to seduce the hens watching the show.



Male sage grouse face off May 1,2020, at a lek in southern Twin Falls County near the Nevada border. Sage grouse rely on vast tracts of sagebrush. Massive wildfires continue to chip away and fragment their habitat, leaving the birds with fewer and smaller areas to forage and breed. Today they occupy just 8% of their historic range.

COLIN TIERNAN, TIMES-NEWS FILE

That charismatic dance is happening less and less as the bird gets attacked from all sides.

Oil, gas and other energy development in several Western states can disrupt leks or chip away at habitat. Invasive species such as cheatgrass, climate change and disease can all hurt sage grouse, too.

In Idaho, wildfire is the main threat. Sage grouse can’t live without vast, unbroken tracts of sagebrush, and big burns in recent decades have removed millions of acres of that habitat. It takes decades for sagebrush to grow back.



A sage grouse hides in its natural sagebrush environment.

Mychel Matthews

Livestock grazing could be hurting sage grouse too — and does in some instances — but the impacts of grazing aren't as clear-cut as the impacts of mining, fire or new development.

Some environmental groups — such as the Western Watersheds Project — say livestock grazing severely degrades sage grouse habitat and is one of the greatest threats facing the struggling species.

Sage grouse biologists generally aren't as sure, often saying grazing is relatively low on the list of threats and emphasizing that its impacts differ greatly from rancher to rancher.



A male sage grouse dances May 1, 2020, at a lek near Rogerson. Biologists say it's fortunate the Badger Fire, when it swept across the South Hills, didn't jump across Deadline Ridge and make its way west into Shoshone Basin, one of Idaho's sage grouse strongholds.

COLIN TIERNAN, TIMES-NEWS FILE PHOTO

Musil emphasized there isn't a lot of good research on the topic.

There's some agreement that nest success and grass height are correlated. Musil said several papers show "the taller the grass the better you're off."

A U.S. Geological Survey, Colorado State University and Utah State University study showed that heavy grazing leads to decreased sage grouse numbers while mild grazing can be beneficial.

"There's times there has been a badly grazed area and it starts a bad cycle," Musil said.

Still, a lot of questions remain about the pros and cons, Musil said. This study will hopefully help answer them, and those answers could guide future grazing management decisions.

Many are eagerly awaiting to see what the grouse and grazing project finds because so much is riding on the sage grouse's fate. If the species ends up on the Endangered Species List — a designation the bird narrowly avoided in 2015 — new federal protections could mean ranchers across the West have to change their operations.

Ranchers and others who operate in the sagebrush may have an economic incentive to keep sage grouse from becoming endangered. If they can graze in a way that's better for grouse, it's probably in their best interest.

Given the stakes, it might seem odd that no one's done a long-term, experimental grazing study before now. Part of the delay might be because of how controversial and emotional the topic is.

"Nobody's had the courage to do it and to do it right," Musil said.

Musil said he never thought a project this big would be possible. It'd require too many collaborators, and ranchers would have to be willing to alter their grazing. Plus, they'd have to be willing to fund and participate in a study that might find something unflattering to their industry.

But ranchers have been willing to help.

"They said, 'No matter what it says we want to know the answer,'" Musil said.



Reese Wray pulls a male sage grouse out of a bag March 18 west of Rogerson.

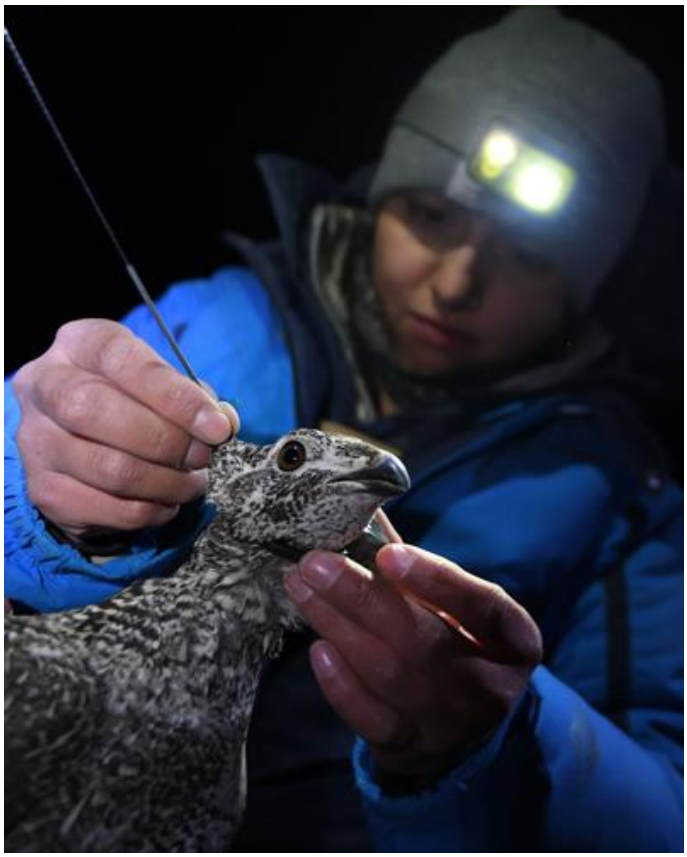
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Simplot holds the grazing permits for the Brown's Bench study site south of Rogerson. Simplot Land Manager Darcy Helmick said this research is incredibly important for the ranching industry.

"I'm confident that whatever we find, what's good for the herd is good for the bird," Helmick said. "We're not scared of what that study's going to show; we're excited to have some science."

Getting closer to the answer

The hen sits calmly in Wray's hands as Siller delicately cinches the radio collar around her neck and bends the antenna against her back. Before letting the bird go, they gently place her in a bag and hoist her from a scale to weigh her.



Wildlife technician Sam Siller makes sure a radio collar fits snugly around a sage grouse hen's neck March 18 west of Rogerson. Siller is part of a capture team collecting data for a massive, 10-year research project looking at the impact of livestock grazing on sage grouse.

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Siller, a wildlife technician, and Wray, an assistant project coordinator, are bundled up in jackets and flannels against the cold. The night is warm, but it can get brutally cold out here in the southern Twin Falls County desert. They don't mind. It's a special opportunity to get to work with sage grouse, they say, and this project's important.



A male sage grouse inflates the yellow air sacs on his chest on May 1, 2020, in Twin Falls County. During the mating season, male sage grouse emit a bizarre pop noise with their air sacs.

TIMES-NEWS FILE

“People might think sage grouse don’t affect them, but they do,” Wray said. “This is our natural resource, our natural heritage.”

Fieldwork for the decade-long project will wrap up in two years. When the data collection’s done the team of biologists will start analyzing the numbers and running statistical analyses in earnest. It’ll be a daunting and slow task to look at so much data and so many variables, Musil said.

At this point, pre-analysis, Musil said it isn’t clear what the study’s going to show, although there’s nothing glaring so far. But he knows many have waited a long time to see what the team finds.

“People are getting really itchy,” Musil said. “They want the answer.”

By Colin Tiernan

Business, agriculture and environment reporter